

Week 1 - (7th March - 14th March)

- Tasks:

- Individual Work:

- **ANDREAS** - Writing up MM documentation and doing Sprint points (and all the UI tasks)
- **MARTIN** - Ensuring communication between the company contact and Neelima, create pitch decks for these meetings
- **MATEUSZ** - Making sure all these tasks are added to git so we can keep track of what needs to be done, in development etc etc
- **GIT ADMIN** (and the tasks given to Jacob and Daniel)
- **AARON** - Communicating with Andreas on the UI elements that must be done.
- **JACOB & DANIEL** - Communicating with the backend team on the tables and if they are appropriate for the data being used by the MLM
- **HUMAIS & DIXON** - Communicating with the UI team on user verification and with the MLM team on if the tables have been made correctly with the correct relationships for the MLM to use.

- UI Team:

- Refine the existing UI so that on the dashboard, it contains a form and an area where the user will be able to see their results and predictions - Andreas is making a list of all the edits that need to be made to the webapp for it to look perfect.
- For user location access, you can take a look at the HTML5 Geolocation API to get users' current locations with their consent. This will require SSL to ensure security.
- Look into implementing email notifications using SMTP protocol or third-party services like SendGrid, integrated via backend APIs, triggered upon the completion of emissions calculations or when the user signs into the app.
- Make sure this is all documented!

- MLM Team:

- Begin gathering datasets related to vehicle emissions, public transport, and geospatial data - can look at producing the artificial data using some sort of python script.
- Use Python for exploratory data analysis (EDA) to understand the structure and quality of the datasets. Python libraries like Pandas for data manipulation and Matplotlib or Seaborn for visualisation will be crucial.
- Initial model considerations could involve looking into regression models for emissions estimation based on transport modes and distances - Look at implementing a vector database using tools like Elastic search for efficient search and matching of commute patterns.
- Make sure this is all documented!

- Backend Team:

- Try and finish up implementing the user verification with the AWS database
- Look into AWS Lambda for serverless API functions that interact with the front end and AI components - how the MLM will
- Look at creating the tables that are outlined in the report given by Martin - Design the database schema focusing on normalisation to reduce redundancy and improve data integrity. This includes creating tables for users, journeys etc with proper foreign keys for relational integrity.
- Make sure this is all documented!

Week 2 - (14th March - 21st March):

- Tasks:

- Individual Work:

- **ANDREAS** - Writing up MM documentation and doing Sprint points (and all the UI tasks)
- **MARTIN** - Ensuring communication between the company contact and Neelima, create pitch decks for these meetings - **ALSO LOOK** at collating information for the final report, so we have enough time at the end to get it written up with all the necessary information and evidence - same marks scheme as last time, look at Andreas's feedback notes from Neelima to get maximise the marks we get - send to Andreas all the work you do and evidence so he can collate it into a folder, easy access when time to write.
- **MATEUSZ** - Making sure all these tasks are added to git so we can keep track of what needs to be done, in development etc etc - **GIT ADMIN** (and the tasks given to Jacob and Daniel)
- **AARON** - Communication with Andreas **AND** the Backend Team to integrate the UI and the Backend to the webapp so user info is stored and can be used by the MLM team in their predictions and calculations.
- **JACOB & DANIEL** - Communicating with Backend team on integrating the ML Models and the tables in the database to fully utilise the tables made in Sprint 1
- **HUMAIS & DIXON** - Communicate with the UI team to try and store the user data from the form into the database - then communicate with the MLM team to connect the ML model with the table so the info the user inputs in can be used by the MLM.

- UI Team:

- Finalising the design of the WebApp, making sure all the connections are seamless and the pages are all to a high quality - Can get feedback from James via sending him a walkthrough of the app and getting his opinion - Document the changes that needed to be made, so we can have evidence and talk about it in the final report!
- Look at trying to implement the forms so that they save the users info they enter into the form to the database - can populate the database with data if needed.
- Martin - Work on data visualisation components, considering libraries like D3.js to display emissions data and potential savings graphically

- MLM Team:

- Make sure that the vector database has been fully implemented
- Continue refining the emissions estimation models and start developing the carpooling suggestion system. This involves more sophisticated machine learning models that could include recommendation systems or neural networks for pattern recognition.
- Try to integrate the MLM to the backend, using the tables created in sprint 1. This could involve creating Python Flask endpoints that the backend can call to process data through the models

- Backend Team:

- Make sure that the User verification/security has been fully implemented
- Make sure the tables have been fully implemented as per the structure that the MLM team require
- Work with the UI Team so that the information that is entered by the user in the form on the dashboard is inputted into the database correctly and safely - no security compromises.
- If possible - work with the MLM team to get the MLM to use the user data stored from the form in the UI - connecting Backend with MLM.

Week 3 - (21st March - 28th March)

- Tasks:

- Individual Tasks:

- **ANDREAS** - Writing up MM documentation and doing Sprint points (and all the UI tasks)
- **MARTIN** - Ensuring communication between the company contact and Neelima, create pitch decks for these meetings - **ALSO LOOK** at collating information for the final report, so we have enough time at the end to get it written up with all the necessary information and evidence - same marks scheme as last time, look at Andreas's feedback notes from Neelima to get maximise the marks we get - send to Andreas all the work you do and evidence so he can collate it into a folder, easy access when time to write.
- **MATEUSZ** - Making sure all these tasks are added to git so we can keep track of what needs to be done, in development etc etc - **GIT ADMIN** (and the tasks given to Jacob and Daniel)
- **AARON** - Try to get user feedback from the webapp and report back to Andreas so the UI team can refine the webapp.
- **JACOB & DANIEL** - Get on with the tasks given below, making sure documentation is written properly on everything so we can include it in the final report. Communication with Martin for Data visualisation is key
- **HUMAIS & DIXON** - Load testing and security testing must be done to test the integrity of the webapp - communication with the MLM team as well to check that the tables and the data are all good.

- UI Team:

- Refine the UI based on initial user feedback, focusing on improving the usability and accessibility of the dashboard.
- Test the front-end for compatibility across different browsers and devices, ensuring a consistent user experience - testing tables & other tests
- POTENTIAL - Implement advanced data visualisation techniques for displaying emissions data and alternative suggestions, possibly integrating libraries like D3.js for interactive charts.
- Write documentation on this!

- MLM Team:

- Finalise the development of the machine learning components, including emission estimations and carpool matching algorithms. Also try and optimise these for accuracy and efficiency.
- Begin integration testing with the backend to validate the machine learning outputs (emission calculations, alternative suggestions, carpool matches) against expected results.
- Can also start looking at Data Visualisation along with Martin - for predicted emission production, per day, per week, per month
- Write documentation on this!

- Backend Team:

- Implement security measures, including encryption for sensitive user data and secure API endpoints to prevent unauthorised access - can talk a lot about this in the report and is a key point.
- Communicating with the MLM team to see if any tables need to be changed, if more data needs to be generated.
- Conduct load testing to ensure the backend can handle expected user volumes, using tools like JMeter or Locust - on the original spec doc it said that the app should have multiple users be able to log in at the same time - can write a testing table for this and show this works
- POTENTIAL - Look at deploying the front-end application to a suitable hosting platform, ensuring it is accessible to users.
- Write documentation on this!

*****Easter Break*** 28th March - 29th April**

Week 4 - (28th March - 4th April)

- WHOLE TEAM:

- Conduct comprehensive testing across the entire platform, including end-to-end testing of user workflows (registration, data submission, viewing results).
- Make a list of all the bugs that are found and try to fix them before the next sprint - Make sure this is all documented!!!

- Writing a Guide on how to use the webapp
- Preliminary ideas on the demo video

- BACKEND TEAM:

- Getting the webapp to be deployed so multiple users can access it at the same time

Week 5 - (4th April - 11th April)

- WHOLE TEAM:

- Somewhat final versions of the Abstract, Main report, software manual and user manual
- Document & code repository finalisation
- More information on these closer to the time - how we will split it up etc

Week 6 - (11th April - 18th April)

- WHOLE TEAM:

- All deliverables should be finished by this time
 - In a state where we can hand them in
- Video can be nearly finished
- Ideas on our stall in the tech showcase - what we can do - how to present the app - powerpoints - interactive elements - how to display each part of our project in a func and cohesive way.

Week 7 - (18th April - 25th April)

- WHOLE TEAM:

- Video, Abstract, Main report, software manual and user manual should be completed by this point.
- Good idea on what our stall will consist of - everyone knows what they need to do for it.
- Double check the webapp is working and can be used on everyone's computers for demonstration purposes.

Week 8 - (25th April - 2nd May)

- WHOLE TEAM:

- Get on with the tasks that have been given from the previous sprint.

Friday 3rd May:

- Abstract
- Main report, software manual and user manual
- Document & code repository (Tag = "PrizeSubmission")
- Demonstration video

Wednesday 15th May:

- Individual report
- Peer assessment

